Mobile and Web-Based Motorcycle Rental Platform for Toba Regency

OUTLINE

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PROBLEM IDENTIFICATION AND SOLUTION

PROBLEM IDENTIFICATION

01 **INEFFICIENCIES IN THE CONVENTIONAL MOTORCYCLE RENTAL PROCESS**

- Customers must visit rental locations in person.
- No guarantee that motorcycles will be available upon arrival.

02 **LACK OF TRANSPARENCY AND COMMUNICATION BETWEEN RENTERS AND OWNERS:**

- Incomplete information about available motorcycles.
- No real-time updates on booking status.

03 **MANUAL DATA MANAGEMENT FOR VENDORS**

- Difficulties in summarizing reports and rental history.

04 **NO CENTRALIZED PLATFORM CONNECTING USERS, VENDORS, AND ADMINS**

SOLUTION

01 **MOBILE APPLICATION FOR USERS**

- Book motorcycles online quickly and conveniently.
- View motorcycle details, booking status, and user reviews.

02 **WEB PLATFORM FOR VENDORS**

- Manage rental listings.
- Monitor and process booking requests in real-time.
- Respond to user reviews.

03 **WEB PLATFORM FOR ADMIN**

- Monitor rental performance and statistics.

CURRENT SYSTEM

The motorbike rental system in Toba is still manual and lacks a centralized digital platform. The current methods include asking locals for rental recommendations, browsing through social media posts, and visiting rental providers directly.

This manual system presents several existing and potential problems:

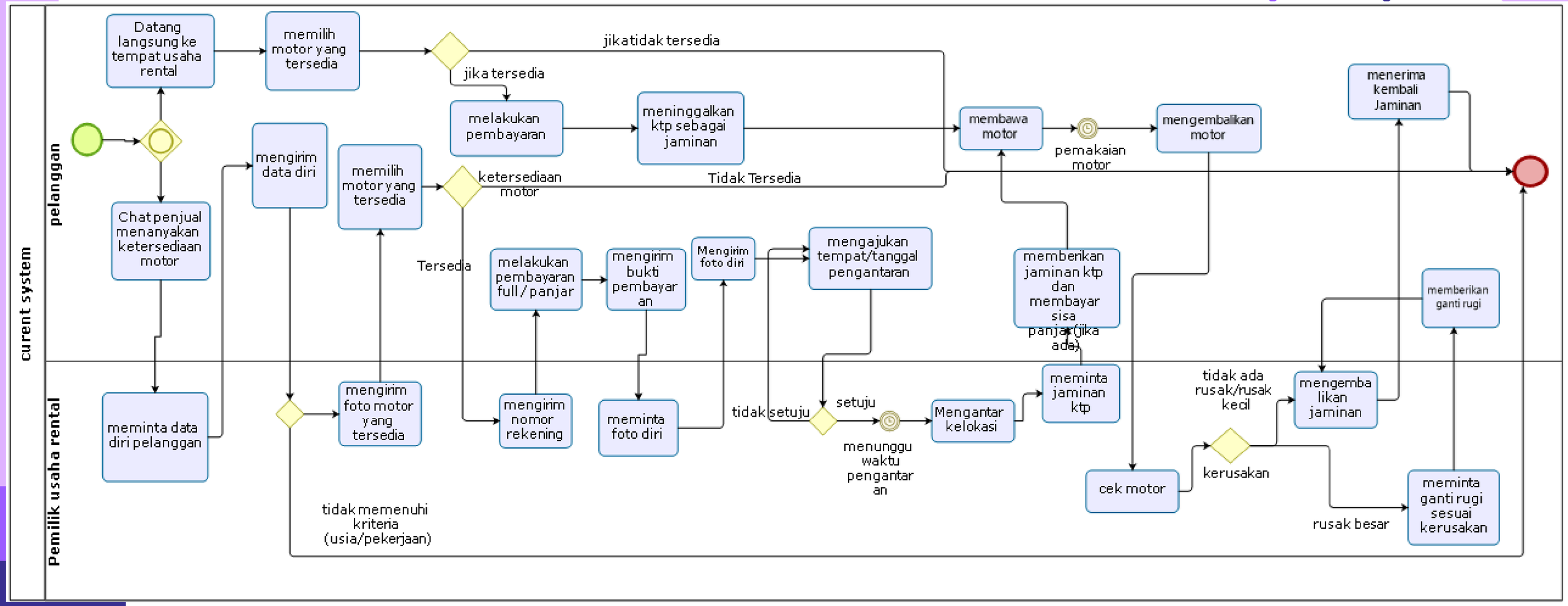
01 Customers cannot easily check motorbike availability in real time, causing delays and uncertainty when trying to rent.

02 Rental owners manage bookings with handwritten notes, making it difficult to track availability and handle rental extensions efficientl

03 Customers must rely on calls or messages, which can be slow or unanswered because there is no automatic notification system for booking updates.

04 There is no way for customers to see feedback or ratings of rental providers, and rental prices vary without a clear standard.

CURRENT SYSTEM



TARGET SYSTEM

MOBILE APLICATION

01 Search and Booking System

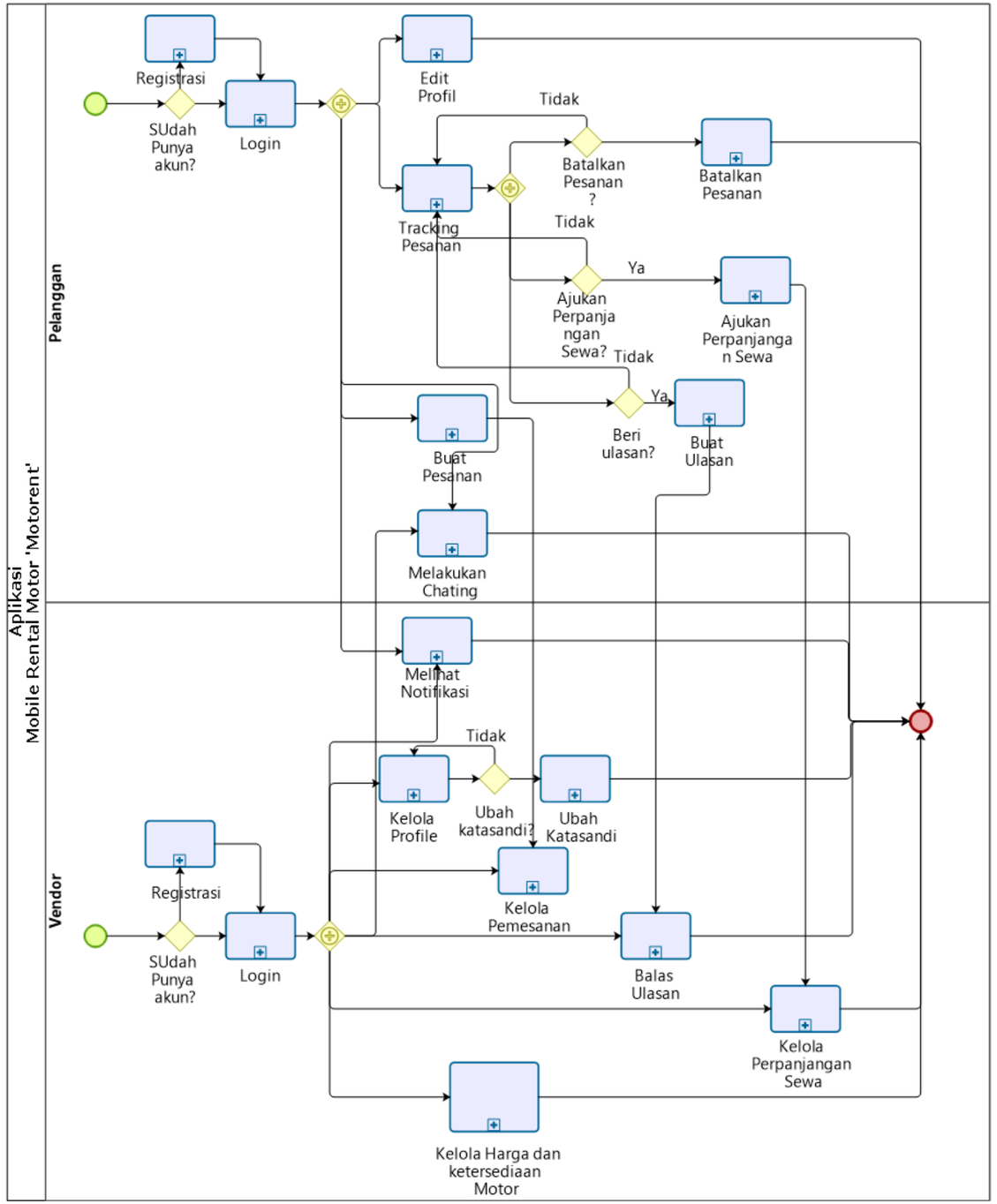
02 Order Management

03 Communication

04 Payment and Additional Services

05 Authentication and Notifications

06 Review and Rating



TARGET SYSTEM

WEBSITE

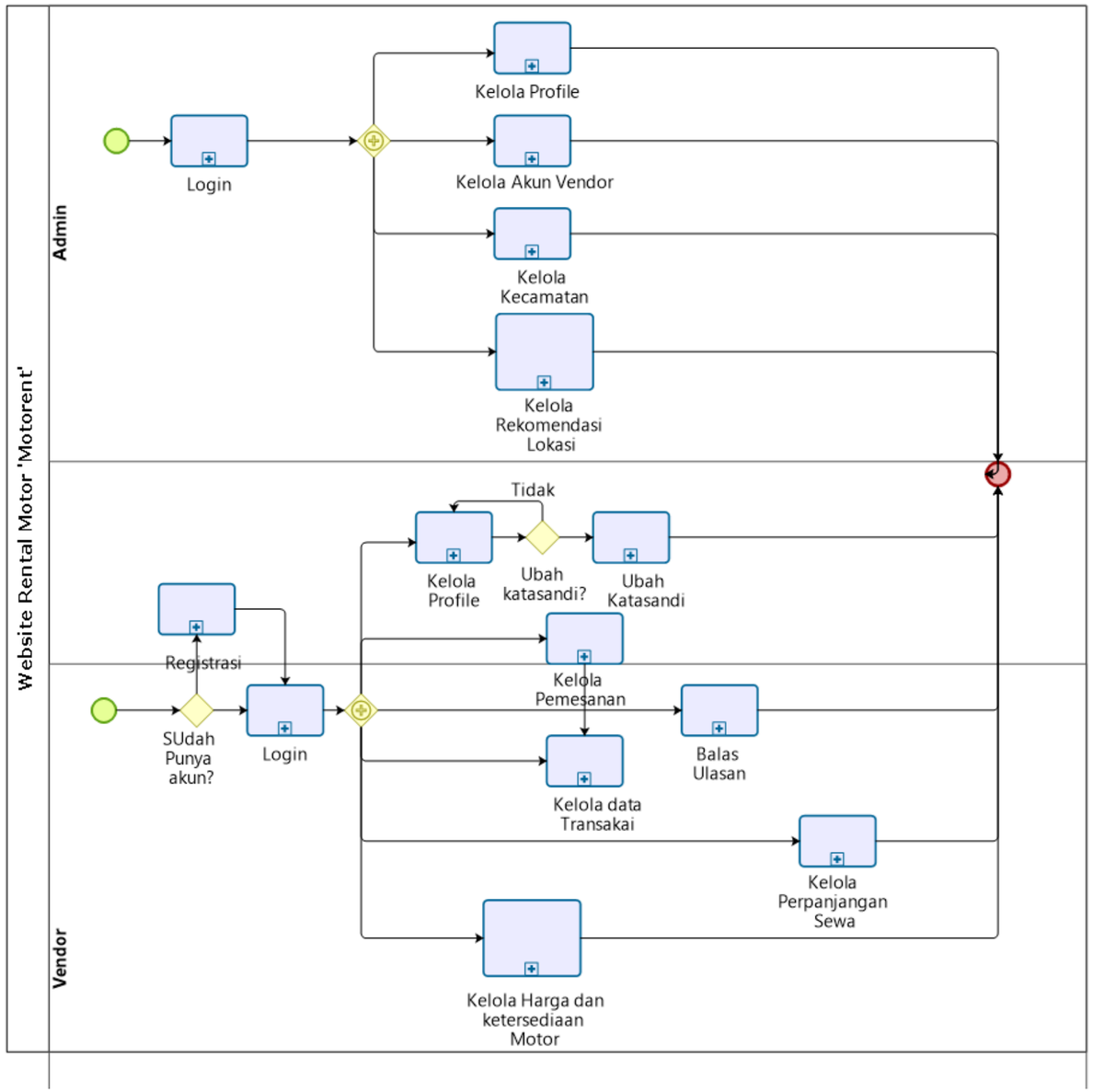
01 Information Access for Customers

02 Business Management for Vendors

03 Order Management for Vendors

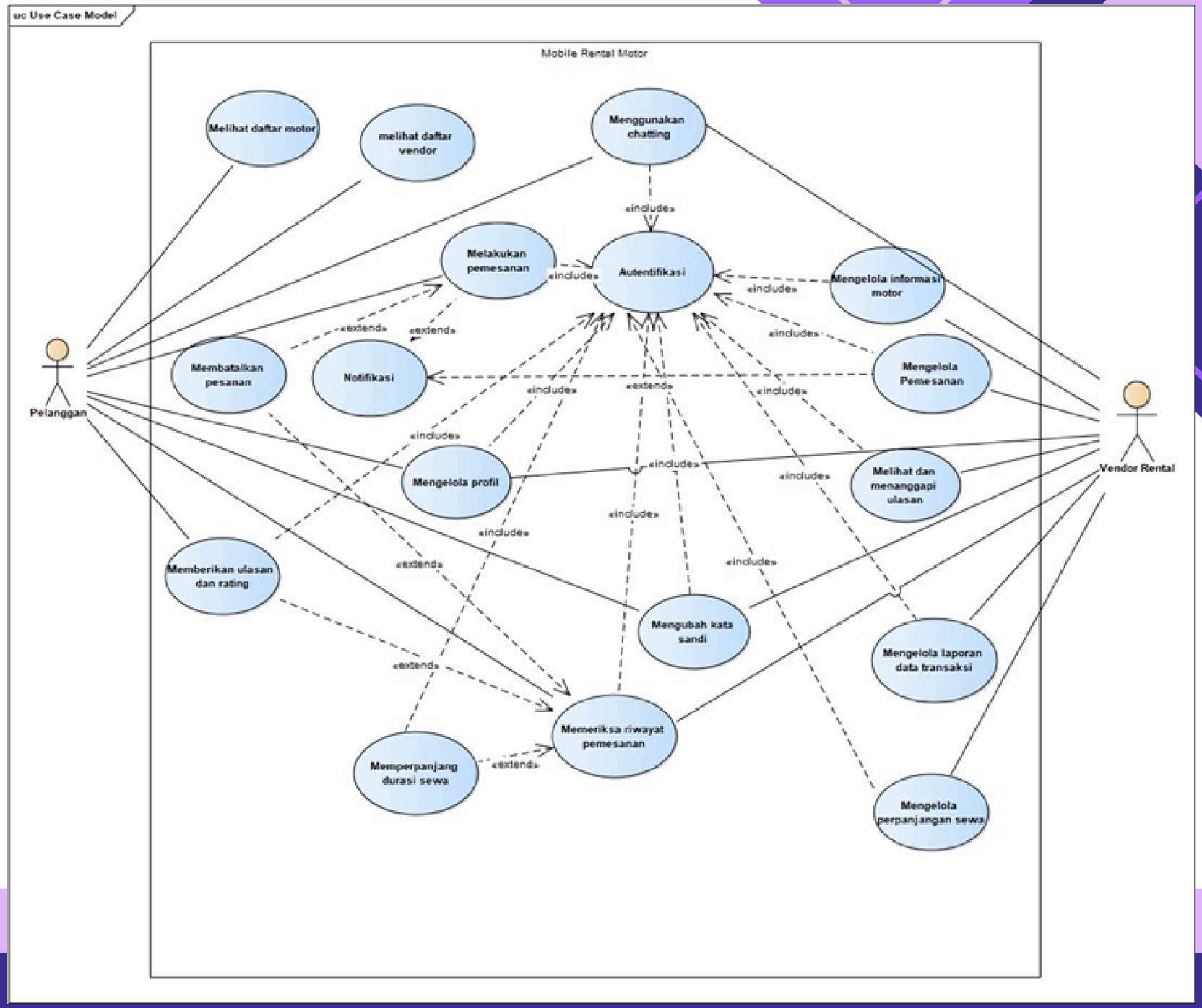
04 Authentication and Verification

05 Administrative Oversight



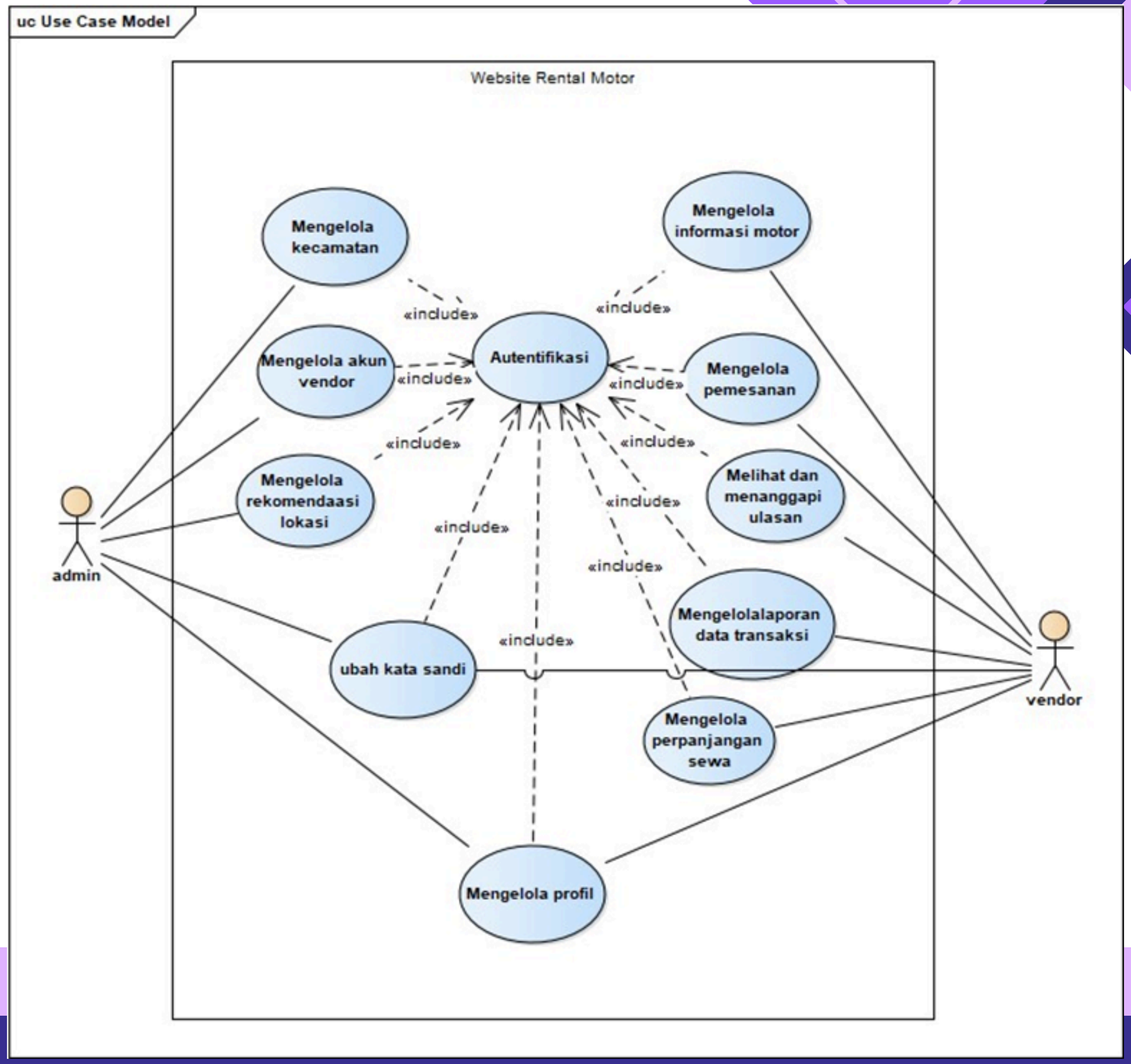
USE CASE DIAGRAM

APPLICATION



USE CASE DIAGRAM

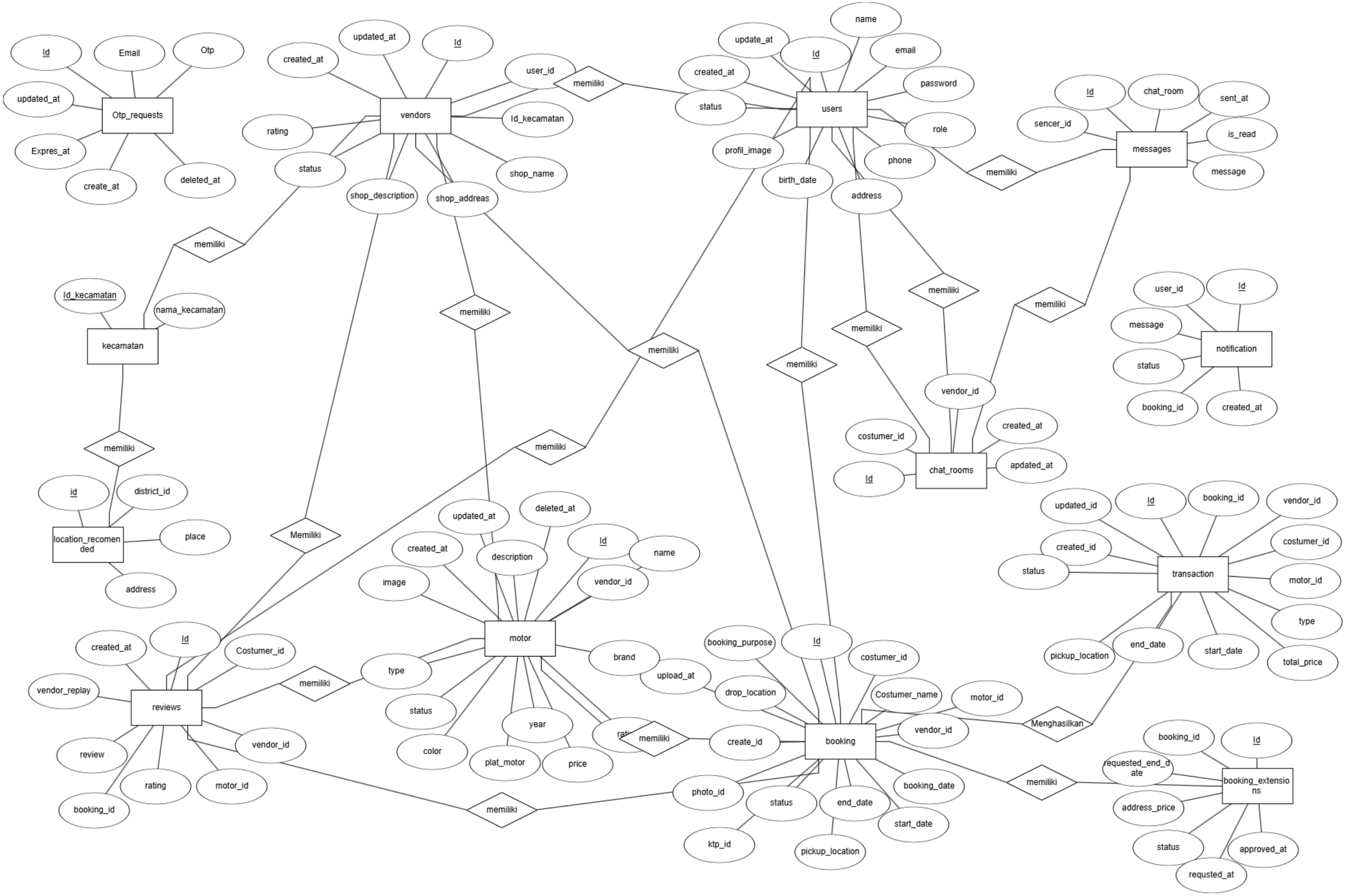
WEBSITE



IMPLEMENTATION

- The mobile application was implemented using **Dart** language with the **Flutter** framework.
- The website was developed using **PHP** language with the **Laravel** framework.
- The server (API) was implemented using **Golang** with the **Gin** framework.

Design ERD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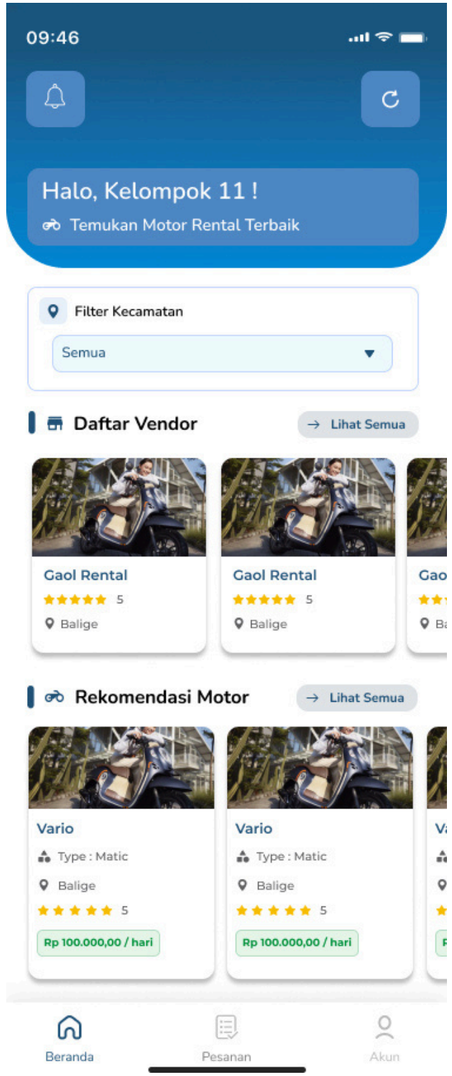
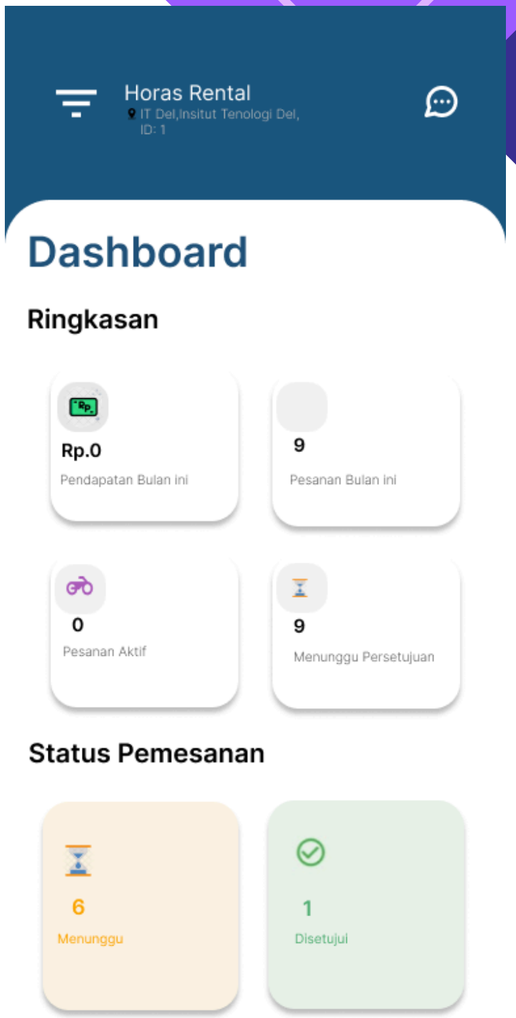
USER INTERFACE FOR MOBILE (FIGMA)



HOMEPAGE USER



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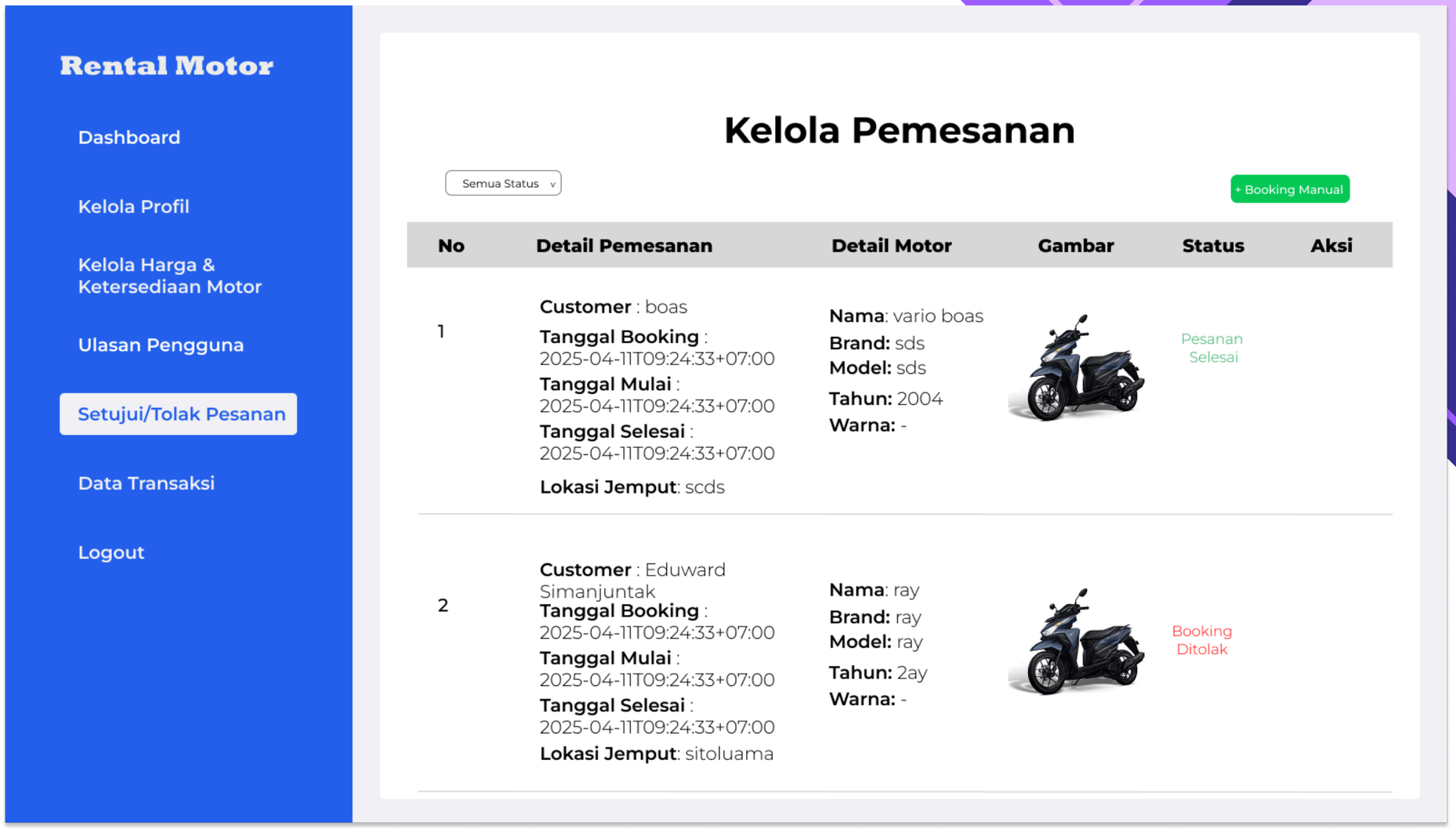
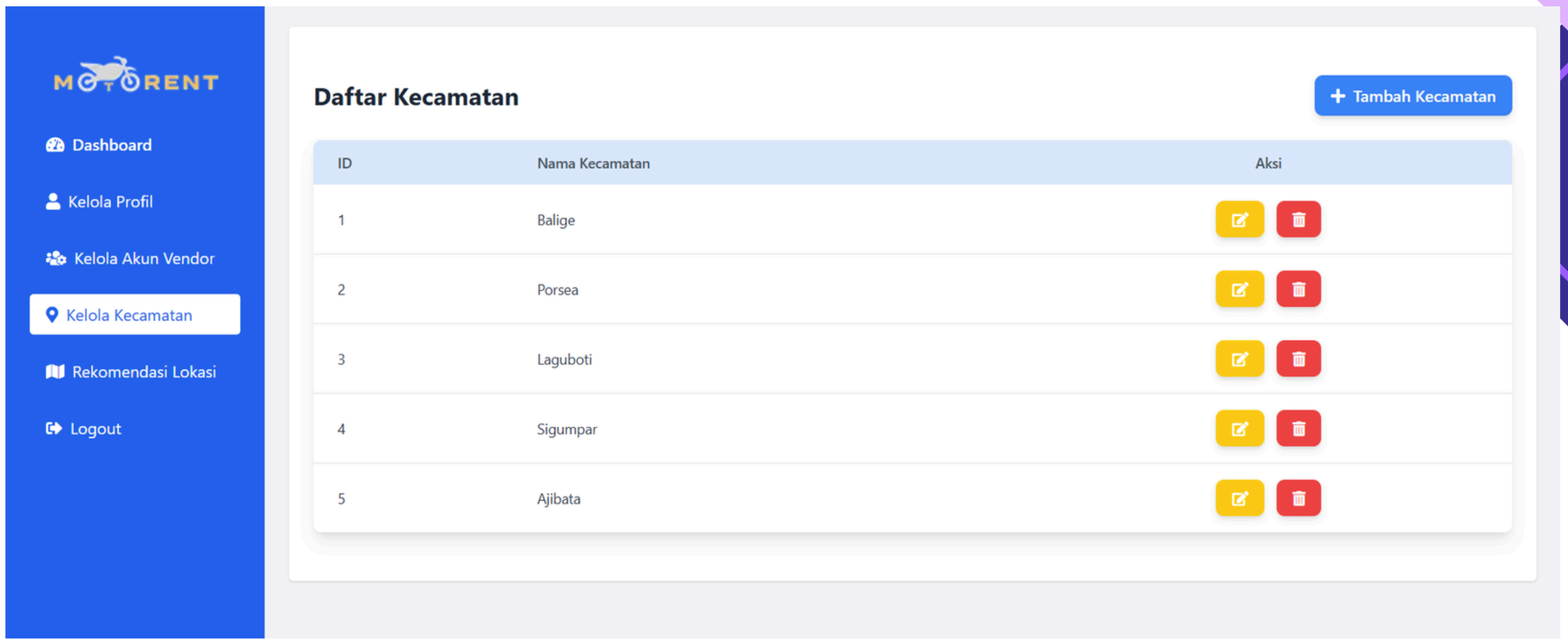
USER INTERFACE FOR WEBSITE (FIGMA)



VENDOR PAGE

USER INTERFACE FOR WEBSITE (FIGMA)



ADMIN PAGE

TESTING

Testing was conducted using two methods to ensure system functionality aligned with the requirements:

- **Selenium testing** was applied to the website for automated UI and functional testing.
- **Manual black box testing** was applied to the mobile application to validate behavior from the user's perspective.
- **Postman** was used to test the server-side API, ensuring each endpoint responds correctly based on test scenarios.

DEPLOYMENT

Website Already released on : https://motorental.d4trpl-itdel.id/

Server Already released on :https://backend-rental-golang-production.up.railway.app/

DEMONSTRATION

Thank You

Kelompok 11

04 Juni, 2025